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United States Patent Application Publication

20250277409

Kind Code

A1

Publication Date

September 04, 2025

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Drilling Motor Bearing Assembly

Abstract

A drilling motor bearing assembly having a longitudinal axis with an outer upper tubular housing threadedly connected to a lower outer tubular housing. The outer upper tubular housing is threadedly engaged to the stator of a power section of a downhole drilling motor assembly. A mandrel having a central bore is threadedly connected to a bearing sub that is threadedly connected to a longitudinally extending rotatable drive shaft so that the mandrel rotates upon rotation of the drive shaft. A lower rotor bearing is connected to the mandrel. A stationary stator bearing is fixed to the lower outer tubular housing at a desired predetermined position so that that the lower rotor bearing rotates with the mandrel relative to the fixed stationary stator bearing. The fixed location of the stator bearing dictates the axial location of the component parts of the bearing assembly.

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Family ID: 96880986

Appl. No.: 18/592439

Filed: February 29, 2024

Publication Classification

Int. Cl.: E21B4/00 (20060101); E21B4/02 (20060101)

U.S. Cl.:

CPC E21B4/003 (20130101); E21B4/02 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] This invention pertains to downhole equipment for oil and gas wells. More particularly, it pertains to a drilling motor bearing assembly for use on a wellbore pipe string such as a coiled tubing string or pipe string and, more particularly, this invention relates to drilling motor bearing assembly which is very robust and easy to assemble.

BACKGROUND OF THE INVENTION

[0002] During the drilling, workover, or plug and abandonment of an oil and gas producing wellbore, a variety of downhole tools may be attached to a pipe or coiled tubing string and utilized to perform various functions within the wellbore. Such downhole tools include downhole drilling motor assemblies having downhole motors used to convert fluid power to rotational power and/or torque. This rotational power is used to rotate a drill bit to create the wellbore in the earth's surface or to re-enter an existing wellbore for remedial operations such as removing scale from the interior surface of a wellbore casing or a tubing string.

[0003] A downhole drilling motor assembly typically includes a power section and a bearing assembly. The power section typically has a housing and a motor used to generate a rotational force to a driveshaft connected to the drill bit. The power section motor may be a positive displacement motor such as a Moineau type power section having a rotor and a stator or the power section motor may be a turbine, a vane motor, or any other devices used to convert fluid power into rotational energy and/or torque. The power section may be comprised of metal, composites, and elastomeric components, or combinations thereof.

[0004] The bearing assembly of the downhole drilling motor assembly is used to support axial thrust loads created from weight on the drill bit, known as on bottom loads, pulling loads from bit overpull, known as off bottom loads, and side or radial loads, such as radial loads generated by a Moineau type power section. If the drilling motor bearing assembly is unable to withstand all of these loads, a catastrophic failure of the downhole drilling motor assembly is likely to occur.

[0005] Drilling motors operate in very demanding environments which include high torques, high pressures, elevated temperatures, abrasive and/or corrosive drilling fluids. The failure of a drilling motor is expensive in both cost and time as the drilling motor must be removed from the wellbore to be replaced. Common drilling motor failures include seal failures which allow lubricant to leak from the drilling motor. Such seal failures may induce premature bearing failure which may result in failed motor mandrels, failed transmission shafts, and seizure of the drilling motor.

[0006] Some drilling motors require inner and outer thrust bearings to be placed into compression during their assembly. A common assembly error is caused when an incorrect compression load is applied to the inner and outer thrust bearings.

[0007] Some drilling motors require spacers in their assembly. Feeler gages, dial calipers, micrometers, and a machine lathe are generally required in order to measure and determine proper spacer lengths and to properly machine spacers to the correct length. Providing a properly sized spacer can be challenging to those with little experience or knowledge of measurement instruments or the operation of a machine lathe.

[0008] Consequently, there is a need for a robust drilling motor that will eliminate the need for seals in order to reduce the incidence of drilling motor failure. There is also a need for a robust drilling motor that eliminates the need to set thrust bearings under a compression load in order to reduce the incidence of error during the drilling motor assembly and the potential for drilling motor failure.

[0009] There is still another need for a robust drilling motor that can be easily manufactured and assembled by inexperienced personnel in order to reduce the potential for assembly errors.

SUMMARY OF THE INVENTION

[0010] The present invention is for an improved drilling motor bearing assembly to be used in conjunction with a Moineau power section of a downhole drilling motor assembly. The proposed

drilling motor bearing assembly eliminates the need for inner and outer thrust bearings and seals and may be assembled without the use of special tools. Because all tolerances are machined into components of the improved drilling motor bearing assembly, errors in assembly due to measuring or machining spacers are virtually eliminated.

[0011] The drilling motor bearing assembly disclosed herein includes an improved means of restraining the stator bearing for both on and off bottom thrust bearings which simplifies the assembly process. The location of the stator bearing is fixed at a desired position within the bearing housing. This fixed static position of the stator bearing relative to adjacent component parts of the bearing assembly dictates the location of all parts axially within the bearing assembly, thus eliminating any possibility of assembly errors. All dimensions of the component parts and the tolerances, clearances, spacing, gaps, and the like between the component parts are predetermined so that the component parts of the drilling motor bearing assembly may be precisely machined prior to assembly. This allows for critical measurements to be taken and machined by experienced machinists during manufacture and remove those responsibilities from the assembly technicians during assembly of the drilling motor bearing assembly.

[0012] When the improved drilling motor bearing assembly is assembled, care must be given only to the torque required to properly makeup the threaded connections of the drilling motor components. Proper application of torque to threaded connections is a necessity in the case threaded connection in all downhole tools and connections, not for just for the threaded connections used in drilling motor bearing assemblies.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a longitudinal cross-sectional view of the drilling motor bearing assembly.

[0014] FIG. 2 is a longitudinal view of the lower rotor bearing affixed to the mandrel.

[0015] FIG. 3 is a longitudinal cross-sectional view of the stator bearing affixed to the bearing housing.

[0016] FIG. 4 is an end view of the stator bearing.

[0017] FIG. 5 is an enlarged partial longitudinal cross-sectional view of the drilling motor bearing assembly taken from Detail 5 of FIG. 1

[0018] FIG. 6 is a longitudinal view of the drilling bearing assembly assembled with the rotor and stator of a power section of a downhole drilling motor assembly attached to a workstring and positioned in a wellbore.

DETAILED DESCRIPTION OF THE INVENTION

[0019] The power section FIG. 1 shows the improved drilling motor bearing assembly 5 utilized in conjunction with a power section (PS) of a downhole drilling motor assembly 200 attached to a pipestring (PS) and placed in a wellbore (WB) as shown in FIG. 6. The motor of the power section (PS) may be comprised of a Moineau type power section having a rotor and a stator or the power section (PS) motor may be comprised of a turbine, a vane motor, or any other devices used to convert fluid power into rotational energy and/or torque.

[0020] Referring to FIG. 1 and FIG. 5, drilling motor bearing assembly 5 has a longitudinal axis 11 along which its component parts are axially arrayed. The drilling motor bearing assembly 5 has an upper end 15, lower end 20, and is comprised of a longitudinally extending outer upper tubular housing 45 threadedly connected to a longitudinally extending lower outer tubular housing 40. The longitudinal axis of the outer upper tubular housing 45 and the lower outer tubular housing 40 are axially aligned with the longitudinal axis 11 of the drilling motor bearing assembly 5.

[0021] The drilling motor bearing assembly 5 has a stator bearing 50 that is fixedly attached to lower tubular housing 40 at a predetermined position along the longitudinal axis the lower tubular

housing **40** such that stator bearing **50** is constrained from both rotation and axial movement relative to lower tubular housing **40**. The attachment of the stator bearing **50** to lower tubular housing **40** may be accomplished by a variety of attachment means including press fitting, brazing, attachment threads, epoxy, glue, or similar techniques. The lower tubular housing **40** and stator bearing **50** may also be constructed as one integral part. The upper tubular housing **45** has a threaded section **120** for threaded engagement to the stator of a power section (PS) and the drive shaft **35** has a threaded section **30** for threaded engagement to the rotor of a power section (PS). [0022] The drilling motor bearing assembly **5** has a longitudinally extending mandrel **10** positioned within the lower tubular housing **40**. Mandrel **10** has a central bore **85** that is axially aligned with the longitudinal axis **11** of the drilling motor bearing assembly **5**. The upper end of the mandrel **10** has a threaded section **100** threadedly connected to the lower end of a bearing sub **55**. The upper end of the bearing sub **55** has a threaded section **110** at its upper end threadedly connected to threaded section **105** at the lower end of a longitudinally extending drive shaft **35**. A lower rotor bearing **60** is connected to the mandrel **10** to correspond with the fixedly connected stator bearing **50** within the lower outer tubular housing **40**. It must be noted that driveshaft **35** and bearing sub **55** may be manufactured in one piece, thereby eliminating a threaded connection. Driveshaft **35** may be comprised of a flex shaft, CV shaft, jaw clutch type shaft, or other suitable torque transmitting shaft.

[0023] FIG. **2** shows the lower rotor bearing **60** threadedly engaged to mandrel **10**. The lower rotor bearing **60** may also be attached to mandrel **10** by other means such as brazing, welding, use of epoxy or glue, and may include castles or splines to prevent relative rotation between the lower rotor bearing **60** and mandrel **10**. Lower rotor bearing **60** and mandrel **10** may also be constructed as one integral part.

[0024] The lower rotor bearing **60** has an upper bearing surface or face **65** that may be comprised of a hard or superhard material to function as a thrust bearing. The hard or superhard material on the upper bearing face **65** of lower rotor bearing **60** may be in the form of buttons or inserts, rings, or discs, or may be applied by a variety of techniques such as a heat treatment process, a coating process, a high velocity oxygen fuel (HVOF) process, or a welding process. In the case of buttons or inserts, a brazing process may be utilized but buttons or inserts may also be pressed into pockets or glued to the surface and combination thereof may be utilized.

[0025] FIG. **3** and FIG. **4** illustrate an embodiment of the stator bearing **50**. As shown in FIG. **4**, an end view of an embodiment of the stator bearing **50**, stator bearing **50** has an upper bearing surface or face **70** comprised of a hard or superhard materials. The upper bearing face **70** may be provided with a plurality of inserts **71** comprised of hard or superhard material such as the semi-circular or arched shaped inserts shown in FIG. **4**. The semi-circular or arched shaped inserts **71** on upper bearing face **70** are for illustration purposes only as the inserts **71** may be circular, ring shaped, pie shaped, or any other shape. Any number of processes to make the upper bearing face **70** of the stator bearing hard or superhard including heat treating, welded on hard metal, hard coatings.

[0026] A hard material is one having a hardness value roughly exceeding 10 gigapascals (GPa) and less than 40 gigapascals (GPa) and a superhard material is one having a hardness value exceeding 40 gigapascals (GPa) when measured using the Vickers Hardness Test. Suitable hard and superhard materials may be comprised of carbide, polycrystalline diamond PDC/PCD, ceramics, or the like.

[0027] The hard or superhard material on upper bearing face **65** of lower rotor bearing **60** is meant to work in conjunction with the hard or superhard material on the lower bearing face **70** or the inserts **71** of stator bearing **50** to function as a thrust bearing. Similarly, hard or superhard material on lower bearing face **80** of bearing sub **55** is meant to work in conjunction with the hard or superhard material on the upper bearing face **75** of stator bearing **50** to function as a thrust bearing. Lower rotor bearing **60** will rotate with mandrel **10** relative to stationary stator bearing **50** about the longitudinal axis of said lower tubular housing and act jointly as “on bottom” thrust bearings.

[0028] In some embodiments the surface or face of one adjacent bearing will be comprised of hard

or superhard materials with adjoining surface or face the corresponding adjacent bearing being simply a polished surface. In other embodiments adjoining bearing surfaces may be comprised of hardened materials but not the same hardened materials. For example, one bearing surface or face may utilize PDC/PCD inserts while the adjoining bearing surface or face may utilize ceramic buttons. The combinations of possibilities is large and must not be confined by those listed here as example.

[0029] As shown in FIG. 3, stator bearing **50** is fixedly positioned at a predetermined location within the lower tubular housing **40**. Fixing a precise location for the stator bearing **50** eliminates the need to set compression on thrust bearings as is the case with many drilling motors. Setting compression on conventional drilling motors having roller ball type thrust bearings or similar bearings is challenging and requires the use of feeler gages, calipers or micrometers, and a machine lathe. In conventional drilling motors, the gap between adjacent surfaces must be carefully measured then a spacer must be accurately machined to properly set compression. Failure to properly set compression leads to premature bearing wear and/or failure. The assembly of conventional drilling motors requires skilled and experienced technicians to properly set compression on thrust bearings.

[0030] Fixing the location of stator bearing **50** of drilling motor bearing assembly **5** at a precise predetermined location within the lower tubular housing **40** of the drilling motor bearing assembly **5** eliminates the need to set compression on the stator bearing **50** during assembly of the drilling motor bearing assembly **5**. The elimination of the need to set compression reduces or substantially eliminates the potential for assembly errors.

[0031] With a fixed location of the stator bearing **50** within the lower tubular housing **40**, the dimension of the component parts of the drilling motor bearing assembly **5** and all tolerances, gaps, and spaces between the component parts of the drilling motor bearing assembly **5** are also predetermined relative to the stationary stator bearing **50**. These predetermined dimensions, tolerances, gaps, and spaces allow the component parts of the drilling motor bearing assembly **5** to be precision machined by experienced machinists skilled in the use of precision measuring instruments such as calipers and micrometers and inspected prior to assembly.

[0032] The fixed location of the stator bearing **50** within the lower tubular housing **40** of the drilling motor bearing assembly **5** also allows the drilling motor bearing assembly **5** to be assembled in only one way. Because the assembly of the drilling motor bearing assembly **5** can be accomplished in only one way, the assembly process is facilitated, even with inexperienced shop technicians, and the potential for incorrect assembly and resulting errors is essentially eliminated.

[0033] FIG. 6 is a view of drilling motor bearing assembly **5** configured with a downhole drilling motor assembly **200**. As shown in FIG. 6, the downhole drilling motor assembly **200** is attached to the workstring (WS) and positioned in the wellbore (WB). The downhole drilling motor assembly **200** includes the power section (PS) and the drilling motor bearing assembly **5** with a drilling bit (BIT) attached at the lower end **20** of the drilling motor bearing assembly **5**. The workstring (WS) may be comprised of a jointed pipe string or coiled tubing string.

[0034] During operation of the downhole drilling motor assembly **200**, drilling fluid enters drilling motor bearing assembly **5** shown in FIG. 1 at its upper end **15** and then travels through annulus **115** between the bore of upper tubular housing **45** and external surface of drive shaft **35**, through ports **90** in bearing sub **55**, into bore **85** of mandrel **10**, then exits lower end **20**. Lower end **20** of the drilling motor bearing assembly **5** is threadedly engaged via threads **25** of the mandrel **10** for rotation of the drilling bit (BIT).

[0035] The drilling bit (BIT) is shown as illustration only. Reamers, junk mills, under reamers, venturi jet baskets, fishing devices, and similar tools meant to be rotated for a desired purpose may be substituted for the drilling bit (BIT) without deviating from the scope of the invention.

Claims

1. A drilling motor bearing assembly comprising: a. an outer upper tubular housing threadedly connected to a lower outer tubular housing; b. a mandrel having a central bore, said mandrel threadedly connected to a bearing sub, said bearing sub having an upper end threadedly connected to a longitudinally extending rotatable drive shaft whereby said mandrel rotates upon rotation of said rotatable drive shaft; c. a lower rotor bearing connected to said mandrel, said lower rotor bearing having an upper rotor bearing face; d. a stationary stator bearing positioned at a predetermined position to said lower outer tubular housing, said stationary stator bearing having a lower stator bearing face; e. a threaded section on said upper tubular housing threadedly engageable to the stator of a power section of a downhole drilling motor assembly; and f. wherein no preloading mechanism is provided.
2. The drilling motor bearing assembly recited in claim 1 wherein: a. said lower stator bearing face on said stationary stator bearing is engageable with said upper rotor bearing face of said lower rotor bearing; and b. wherein said lower rotor bearing rotates with said mandrel relative to said stationary stator bearing.
3. The drilling motor bearing assembly recited in claim 2 wherein said stationary stator bearing is constrained from rotation and from axial movement relative to said lower outer tubular housing.
4. The drilling motor bearing assembly recited in claim 3 wherein said lower stator bearing face on said stationary stator is comprised of a hard or super hard material.
5. The drilling motor bearing assembly recited in claim 4 wherein said upper rotor bearing face on said lower rotor bearing is comprised of a hard or superhard material
6. The drilling motor bearing assembly recited in claim 5 further comprising a workstring connected to said power section of said downhole drilling motor assembly with said threaded section on said outer upper tubular housing threadedly engaged to said stator of said power section of said downhole drilling motor assembly.
7. The drilling motor bearing assembly recited in claim 6 wherein said hard or superhard material of said stator upper bearing face of said stationary stator and wherein said hard or superhard material of said upper rotor bearing face of said lower rotor bearing are selected from the group comprised of carbide, polycrystalline diamond PDC/PCD, or ceramics.
8. The drilling motor bearing assembly recited in claim 6 wherein said stationary stator bearing is integral with said lower outer tubular housing.
9. A drilling motor bearing assembly, said drilling motor bearing assembly having component parts comprising: a. an outer upper tubular housing threadedly connected to a lower outer tubular housing along a longitudinal axis; b. a mandrel having a central bore, said mandrel threadedly connected to a bearing sub, said bearing sub having an upper end threadedly connected to a longitudinally extending rotatable drive shaft whereby said mandrel rotates upon rotation of said rotatable drive shaft; c. a lower rotor bearing connected to said mandrel; d. a stationary stator bearing fixed to said lower outer tubular housing at a desired predetermined axial position within said lower outer tubular housing wherein said stationary stator bearing is constrained from rotation and from axial movement relative to said lower outer tubular housing; e. a threaded section on said upper tubular housing threadedly engageable to the stator of a power section of a downhole drilling motor assembly; and f. wherein no preloading mechanism is provided
10. The drilling motor bearing assembly recited in claim 9 wherein said lower rotor bearing rotates with said mandrel relative to said fixed stationary stator bearing.
11. The drilling motor bearing assembly recited in claim 10 wherein said predetermined location of said stationary stator bearing dictates the axial location of said component parts of said drilling motor bearing assembly along said longitudinal axis.
12. The drilling motor bearing assembly recited in claim 11 wherein said desired predetermined

axial position of said stationary stator bearing within said lower outer tubular housing predetermines the tolerances, gaps, and spacing between said component parts of said drilling motor bearing assembly.

13. The drilling motor bearing assembly recited in claim 12 wherein said lower outer tubular housing is integral with said stationary stator bearing.

14. The drilling motor bearing assembly recited in claim 13 wherein said driveshaft is an integral component of said bearing sub.

15. The drilling motor bearing assembly recited in claim 14 further comprising a workstring connected to a power section of said downhole drilling motor assembly, said power section having a stator and a rotor, said threaded section on said outer upper tubular housing threadedly engaged to said stator of said power section.

16. The drilling motor bearing assembly recited in claim 14 wherein said desired predetermined axial position of said stationary stator bearing allows said drilling motor bearing assembly to be assembled only one way.

17. A downhole drilling motor assembly comprising: a. workstring; b. a power section having a rotor connected to a longitudinally extending rotatable driveshaft; and c. a bearing assembly, said bearing assembly having: i. a longitudinally extending outer upper tubular housing threadedly connected to a longitudinally extending lower outer tubular housing along a longitudinal axis; ii. a mandrel having a central bore, said mandrel threadedly connected to a bearing sub, said bearing sub having an upper end threadedly connected to a longitudinally extending rotatable drive shaft whereby said mandrel rotates about said longitudinal axis of said lower tubular housing upon rotation of said rotatable drive shaft; iii. a lower rotor bearing connected to said mandrel; iv. a stationary stator bearing integral with said lower outer tubular housing at a desired predetermined axial position, said stationary stator bearing constrained from rotation and axial movement relative to said lower outer tubular housing; v. a threaded section on said upper tubular housing threadedly engageable to said stator of said power section; vi. wherein said lower rotor bearing of said bearing assembly rotates with said mandrel relative to said fixed stationary stator bearing about said longitudinal axis of said lower tubular housing; and vii. wherein no preloading mechanism is provided.

18. The downhole drilling motor assembly recited in claim 17 wherein said desired predetermined axial position of said stationary stator bearing within said lower tubular housing of said bearing assembly predetermines the axial location of the component parts of said bearing assembly along said longitudinal axis and predetermines the tolerances, gaps, and spacing between said component parts of said bearing assembly.

19. The downhole drilling motor assembly recited in claim 18 wherein said stationary stator bearing of said bearing assembly has a lower stator bearing face engageable with an upper rotor bearing face on said lower rotor bearing and wherein said lower stator bearing face and said upper rotor bearing face of said bearing assembly are comprised of hard or super hard material selected from the group comprised of carbide, polycrystalline diamond PDC/PCD, or ceramics.

20. The downhole drilling motor assembly recited in claim 19 wherein said driveshaft is an integral component of said bearing sub.

21. A method for assembly of a downhole drilling motor bearing comprising the steps of: a. providing a bearing assembly having component parts comprising: i. a longitudinally extending outer upper tubular housing threadedly connected to a longitudinally extending lower outer tubular housing along a longitudinal axis; ii. a mandrel having a central bore; iii. a lower rotor bearing connected to said mandrel; iv. a stationary stator bearing integral with said lower outer tubular housing at a desired predetermined axial position, said stationary stator bearing constrained from rotation and axial movement relative to said lower outer tubular housing; v. a threaded section on said stationary stator; vi. a threaded section on said upper tubular housing; b. threadedly engaging said threaded section on said stationary stator with said threaded section of said power section

without compression whereby said lower rotor bearing of said bearing assembly rotates with said mandrel relative to said fixed stationary stator bearing about said longitudinal axis of said lower tubular housing; and c. wherein no preloading mechanism is provided.

22. The method for assembly of a downhole drilling motor bearing recited in claim 21 wherein said step of providing said bearing assembly includes fixing said stationary stator bearing of said bearing assembly at a desired predetermined location within said lower tubular housing of said bearing assembly thereby predetermining the tolerances, gaps, and spacing between said component parts of said bearing assembly.

23. The method for assembly of a downhole drilling motor bearing recited in claim 22 comprising the additional steps of: a. providing a workstring having a power section, said power section having a rotor connected to a longitudinally extending rotatable driveshaft; and b. providing a bearing sub; c. connecting said bearing sub to said mandrel of said bearing assembly, d. connecting said bearing sub to said longitudinally extending rotatable drive shaft whereby said mandrel of said bearing assembly rotates about said longitudinal axis of said lower tubular housing of said bearing assembly upon rotation of said rotatable drive shaft.

24. The method for assembly of a downhole drilling motor bearing recited in claim 23 wherein said stationary stator bearing of said bearing assembly has a lower stator bearing face engageable with an upper rotor bearing face on said lower rotor bearing and wherein said lower stator bearing face and said upper rotor bearing face are comprised of hard or super hard material selected from the group comprised of carbide, polycrystalline diamond PDC/PCD, or ceramics.

25. The method for assembly of a downhole drilling motor bearing recited in claim 24 wherein said driveshaft is an integral component of said bearing sub.
